

1. Tail control

1. Sample mean and target

For independent observations $Z_1, \dots, Z_n$ with population mean $\mu = E[Z]$,

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n Z_i.$$

A finite-sample deviation guarantee has the form

$$P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq \delta.$$

Always identify the event, tolerance ε , confidence parameter δ , and assumptions before applying a bound.

2. Markov and Chebyshev

Markov. If $X \geq 0$ and $a > 0$, then

$$P(X \geq a) \leq \frac{E[X]}{a}.$$

Chebyshev. If X has finite mean μ and variance σ^2 , then for $\varepsilon > 0$,

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

For i.i.d. variables with variance σ^2 ,

$$\text{Var}(\hat{\mu}_n) = \frac{\sigma^2}{n}, \quad P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

The $\frac{\sigma^2}{n}$ step uses independence.

3. Hoeffding inequality

If independent X_i satisfy $a_i \leq X_i \leq b_i$ almost surely, then for $t > 0$,

$$P\left(\sum_{i=1}^n (X_i - E[X_i]) \geq t\right) \leq \exp\left(-2 \frac{t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Apply the result to both signs for a two-sided bound. For independent $Z_i \in [0, 1]$,

$$P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq 2 \exp(-2n\varepsilon^2).$$

To ensure the right side is at most δ ,

$$n \geq \frac{\ln(\frac{2}{\delta})}{2\varepsilon^2}.$$

This is a sufficient sample size, not necessarily the exact minimum.

4. Which bound is justified?

- **Mean only + nonnegative variable:** Markov.
- **Finite variance:** Chebyshev.
- **Known bounded intervals + independence:** Hoeffding.

For a sample mean, Chebyshev decays as $\frac{1}{n}$; Hoeffding gives exponential decay in n under stronger assumptions.

Pitfall: Bounded time-series losses do not justify the displayed Hoeffding bound unless independence (or another suitable dependence result) is established.

2. Finite-dimensional geometry

1. L^p norms

For $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ and $1 \leq p < \infty$,

$$\|x\|_p = \left(\sum_{j=1}^d |x_j|^p\right)^{\frac{1}{p}}, \quad \|x\|_\infty = \max_{1 \leq j \leq d} |x_j|.$$

Useful cases:

- $\|x\|_1 = \sum_j |x_j|$
- $\|x\|_2 = \sqrt{\sum_j x_j^2}$
- $\|x\|_\infty = \max_j |x_j|$

If $1 \leq p \leq q \leq \infty$,

$$\|x\|_q \leq \|x\|_p \leq d^{\frac{1}{p} - \frac{1}{q}} \|x\|_q.$$

Fixed-dimensional norms induce the same convergence, but their constants and level-set shapes are not interchangeable. For $0 < p < 1$, the displayed quantity can fail the triangle inequality and is not called a norm here.

2. Inner product and Cauchy-Schwarz

For $x, y \in \mathbb{R}^d$,

$$\langle x, y \rangle = \sum_{j=1}^d x_j y_j.$$

The Euclidean norm is induced by the inner product: $\|x\|_2 = \sqrt{\langle x, x \rangle}$.

Cauchy-Schwarz. For all $x, y \in \mathbb{R}^d$,

$$|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2.$$

Equality holds exactly when x and y are linearly dependent, including zero-vector cases. For $y \neq 0$, choosing $t = \frac{\langle x, y \rangle}{\|y\|_2^2}$ in $0 \leq \|x - ty\|_2^2$ gives the squared inequality.

3. Geometry changes optimization

For the score $2x_1 + x_2$:

- under $\|x\|_1 \leq 1$, the maximum is 2 at $x = (1, 0)$, a sparse vector;
- under $\|x\|_2 \leq 1$, Cauchy-Schwarz gives maximum $\sqrt{5}$ at $x = \frac{2,1}{\sqrt{5}}$, a diffuse vector.

Changing the norm can therefore change the optimizer's structure even though norms are equivalent in fixed dimension.

3. Functions as vectors

1. The space $L^2([0, 1])$

Elements are real functions satisfying

$$\int_0^1 |f(t)|^2 dt < \infty,$$

where functions equal almost everywhere are treated as the same element. Define

$$\langle f, g \rangle = \int_0^1 f(t)g(t) dt, \quad \|f\|_2 = \sqrt{\int_0^1 |f(t)|^2 dt}.$$

Addition and scalar multiplication are pointwise, up to almost-everywhere equality. The same Cauchy-Schwarz bound holds with these definitions.

Hilbert space. An inner-product space H is Hilbert when it is complete in $\|h\| = \sqrt{\langle h, h \rangle}$: every norm-Cauchy sequence converges in norm to an element still in H . The space $L^2([0, 1])$ is Hilbert. Completeness prevents a norm-limit from leaving the chosen space.